

MECHANICAL DESIGN AND SIMULATION OF THE ATTITUDE DETERMINATION AND CONTROL SYSTEM IN A 3U CUBESAT MODEL

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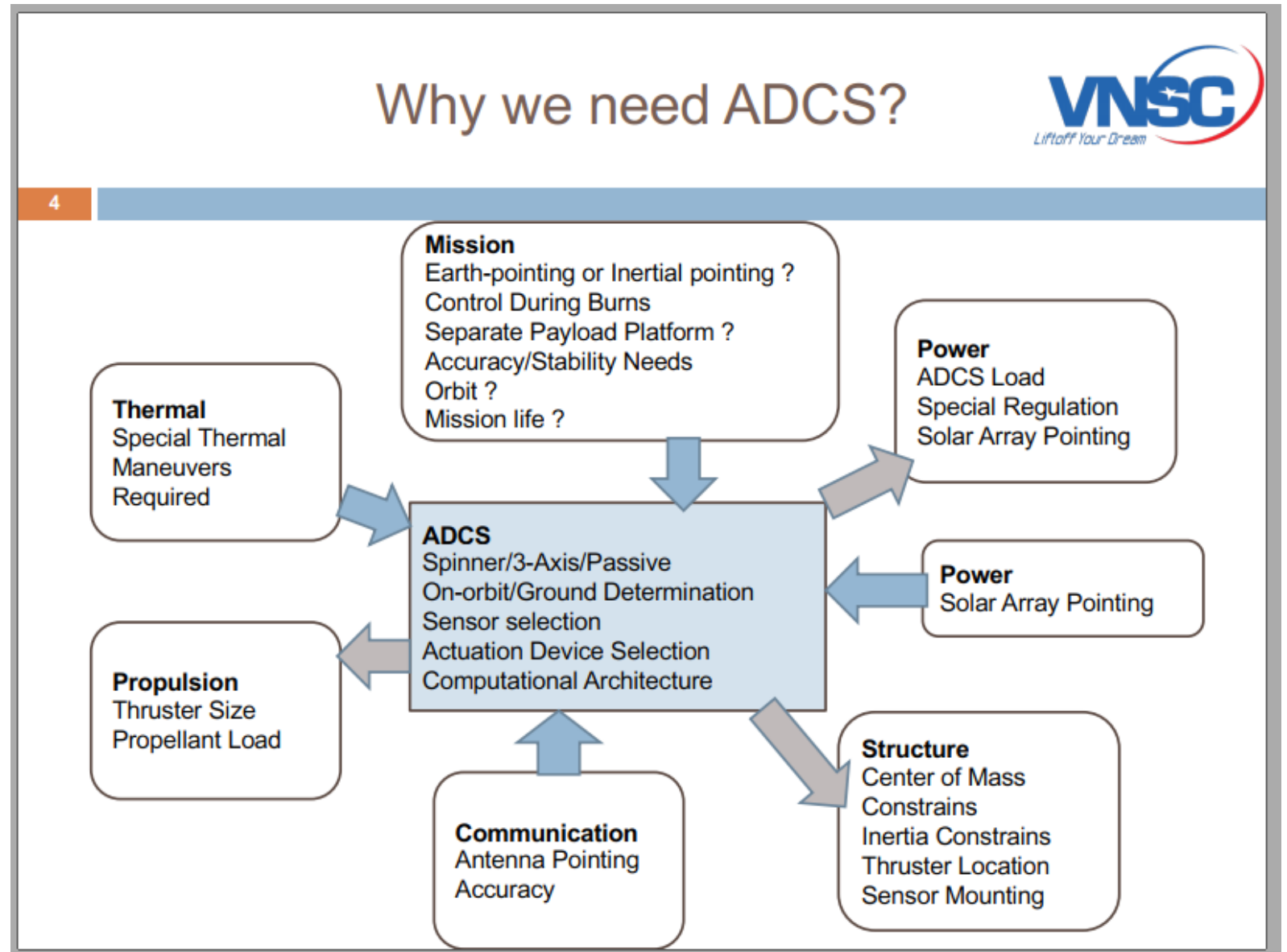


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1. Introduction/ Literature Review

- CubeSats are developed worldwide, not only they are served the education purpose but they also can perform science tasks when launched into space.
- -> Development of a 3U CubeSat model in USTH.
- The ADCS is a vital system for a spacecraft. The simulation of ADCS is needed for visualizing the motion of the spacecraft.

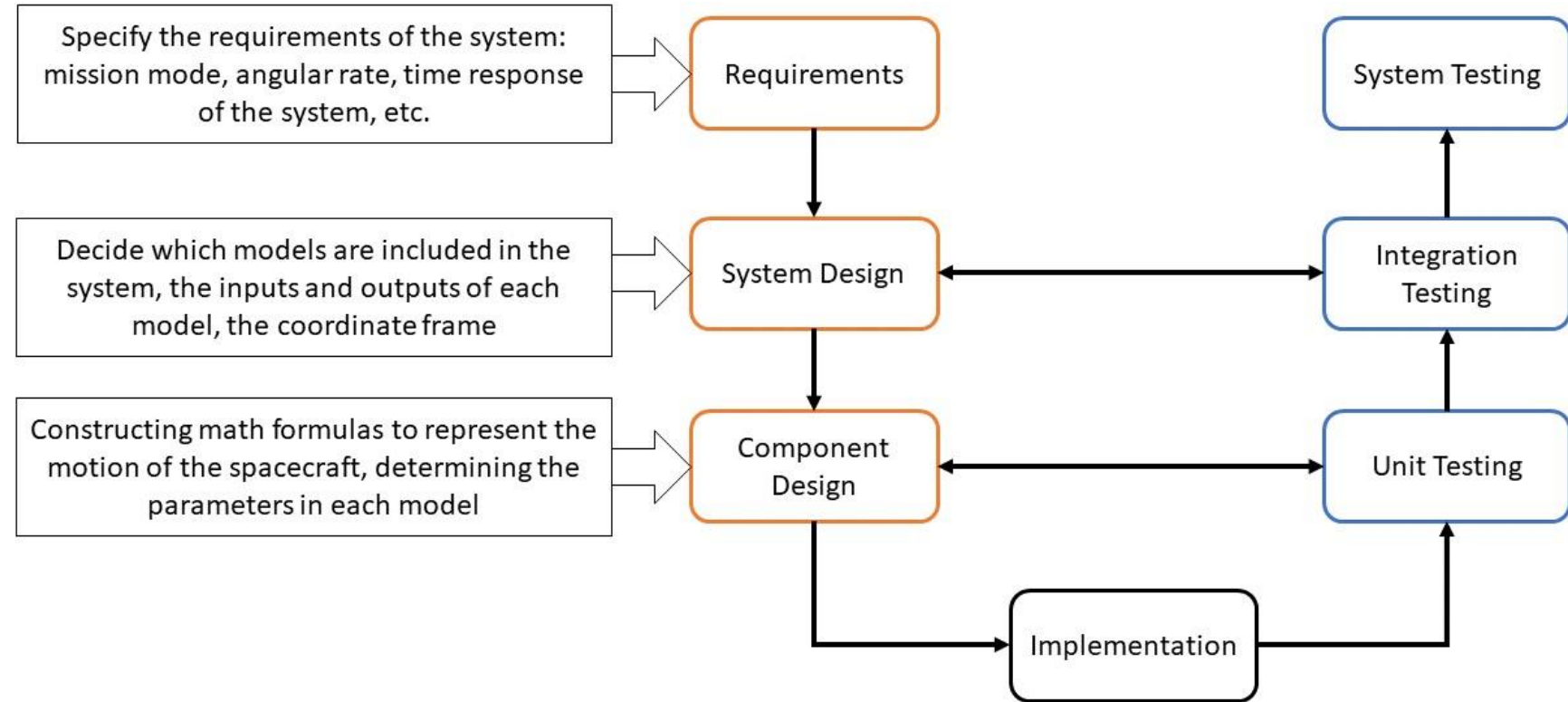


Source: Le, H.,(2020). *Lectures in "Space System Design" module*, USTH

2. Requirements and Tools

2.1. Requirements:

The CubeSat should spin with the rate between 0.1 and 1 degree per second, and pointing with the accuracy less than 5 degrees.



2.2.Tools:



Aerospace Toolbox

R2021a by MathWorks

Analyze and visualize aerospace vehicle motion using reference standards and models



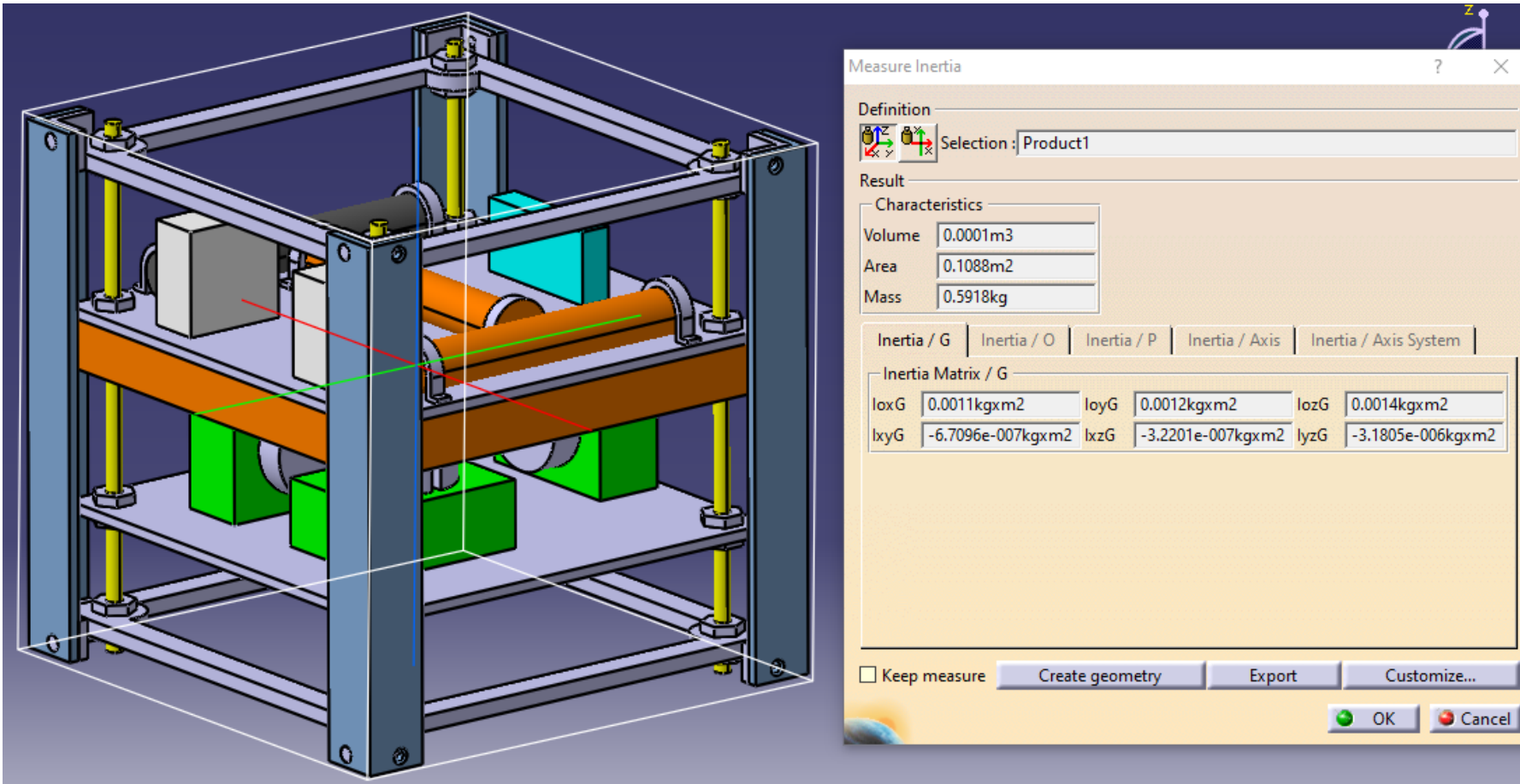
DSP System Toolbox

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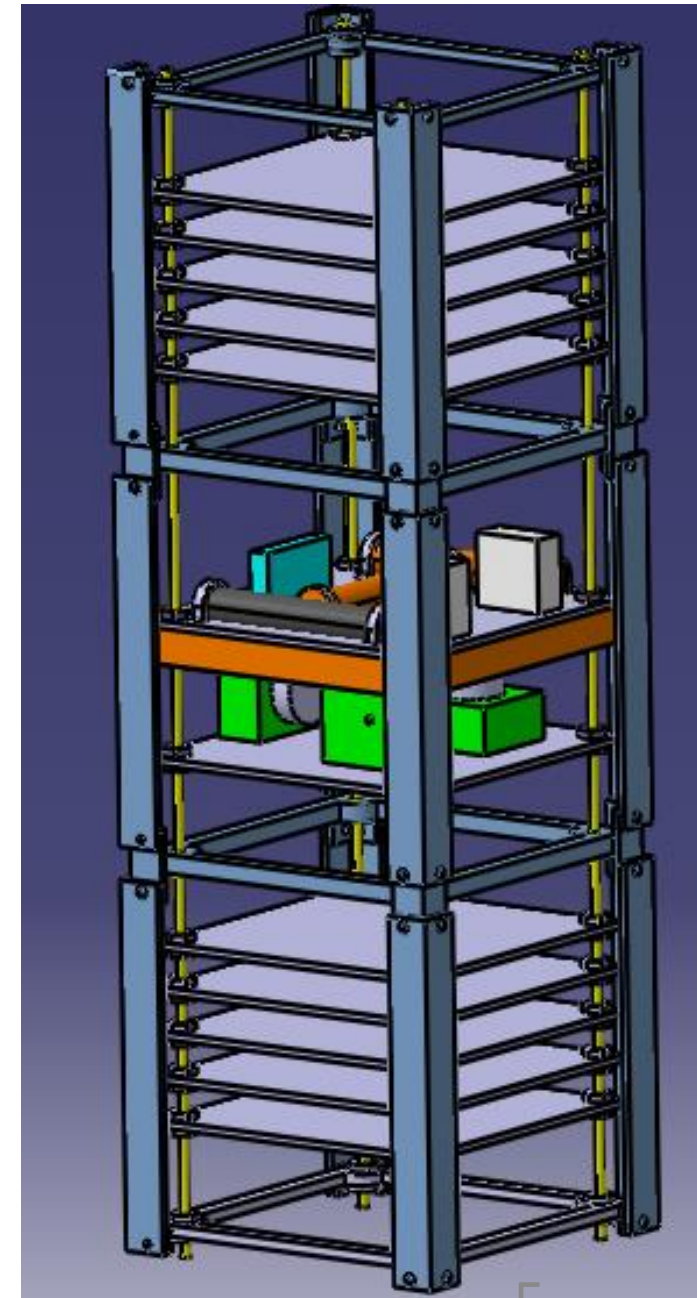
Design and simulate streaming signal processing systems



3. Mechanical design on CATIA

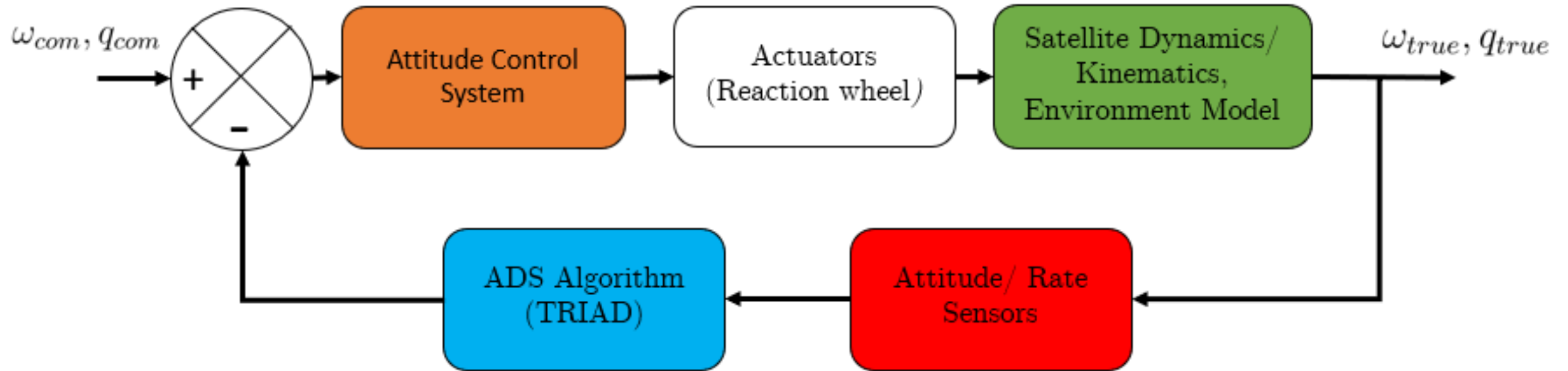


$$I = \begin{bmatrix} 0.0011 & 0 & 0 \\ 0 & 0.0012 & 0 \\ 0 & 0 & 0.0014 \end{bmatrix} \text{ kg.m}^2$$



4. MATLAB/Simulink Simulation

The block diagram of the ADCS:

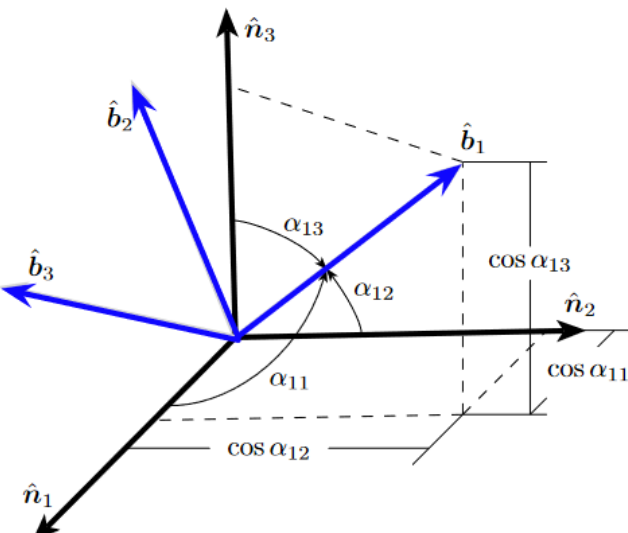


Decisions on rotation frames, sensors and algorithm:

- Sensors: Sun sensors, Gyroscope and Magnetic sensors
- ADS algorithm: TRIAD

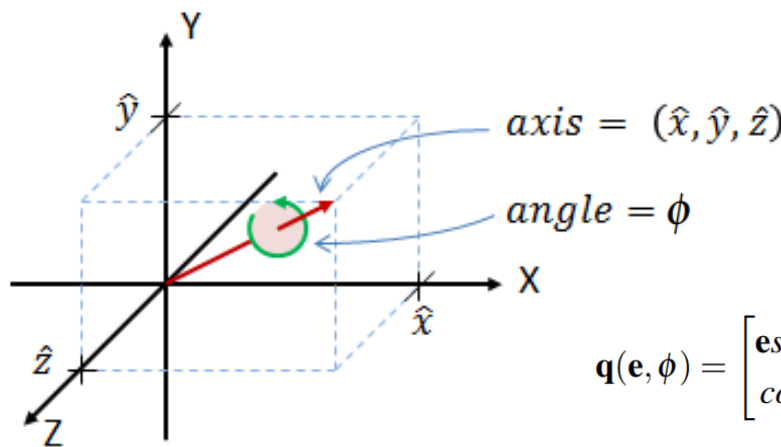
4.1. Attitude Representations and coordinate frames

Direction Cosine Matrix (DCM)



$$\{\hat{\mathbf{b}}\} = \begin{bmatrix} \cos\alpha_{11} & \cos\alpha_{12} & \cos\alpha_{13} \\ \cos\alpha_{21} & \cos\alpha_{22} & \cos\alpha_{23} \\ \cos\alpha_{31} & \cos\alpha_{32} & \cos\alpha_{33} \end{bmatrix} \{\hat{\mathbf{n}}\} = [\mathbf{A}]\{\hat{\mathbf{n}}\}$$
$$\{\hat{\mathbf{n}}\} = [\mathbf{A}]^T \{\hat{\mathbf{b}}\}$$

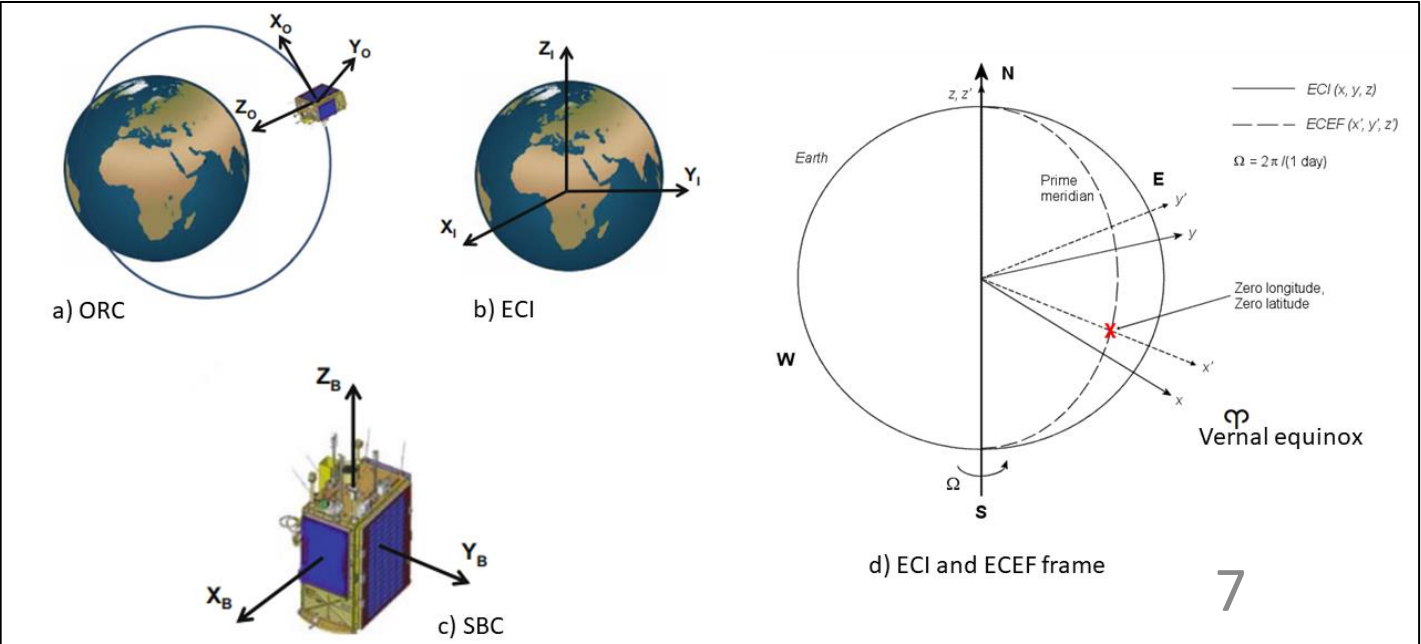
Quaternion



$$\mathbf{q}(\mathbf{e}, \phi) = \begin{bmatrix} \mathbf{e} \sin(\phi/2) \\ \cos(\phi/2) \end{bmatrix} = \begin{bmatrix} e_1 \sin(\phi/2) \\ e_2 \sin(\phi/2) \\ e_3 \sin(\phi/2) \\ \cos(\phi/2) \end{bmatrix} = \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{bmatrix}$$

Coordinate frames

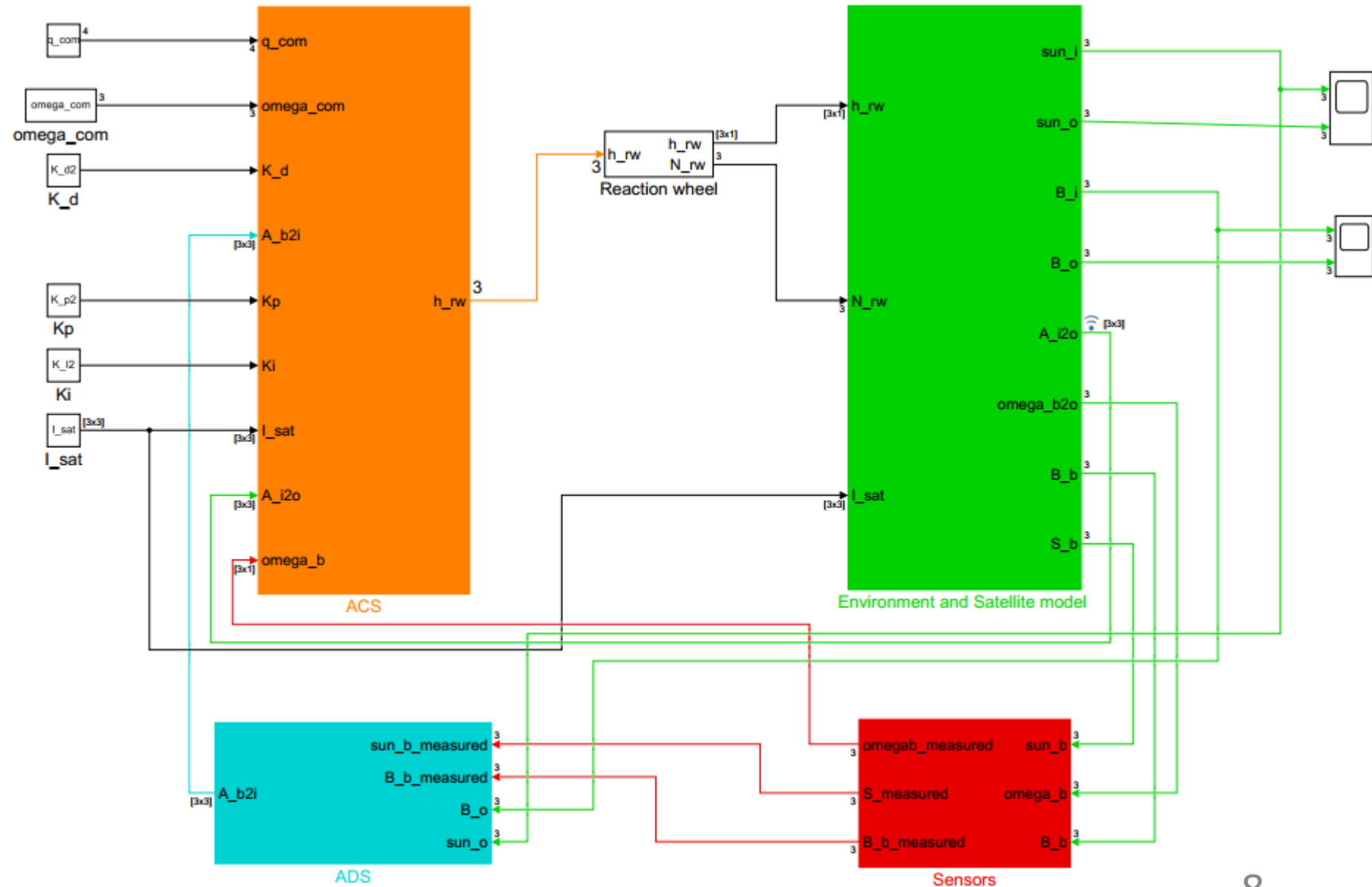
- a) Orbit Reference Coordinate (ORC)
- b) Earth-centered Inertial Coordinate (ECI)
- c) Spacecraft Body Coordinate (SBC)
- d) Earth-Center/Earth-Fixed Coordinate (ECEF)

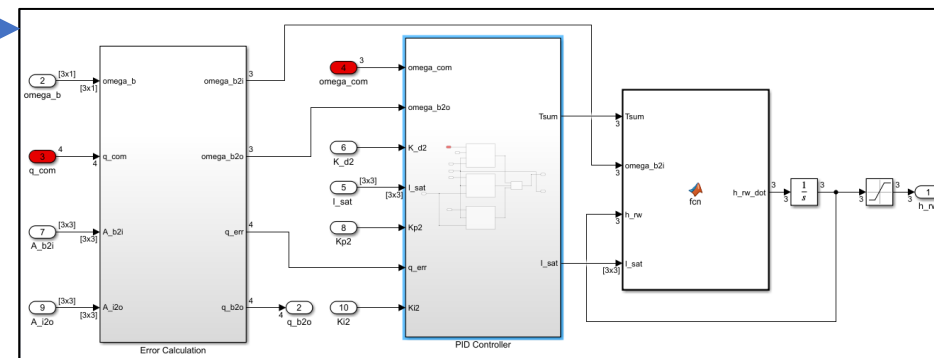
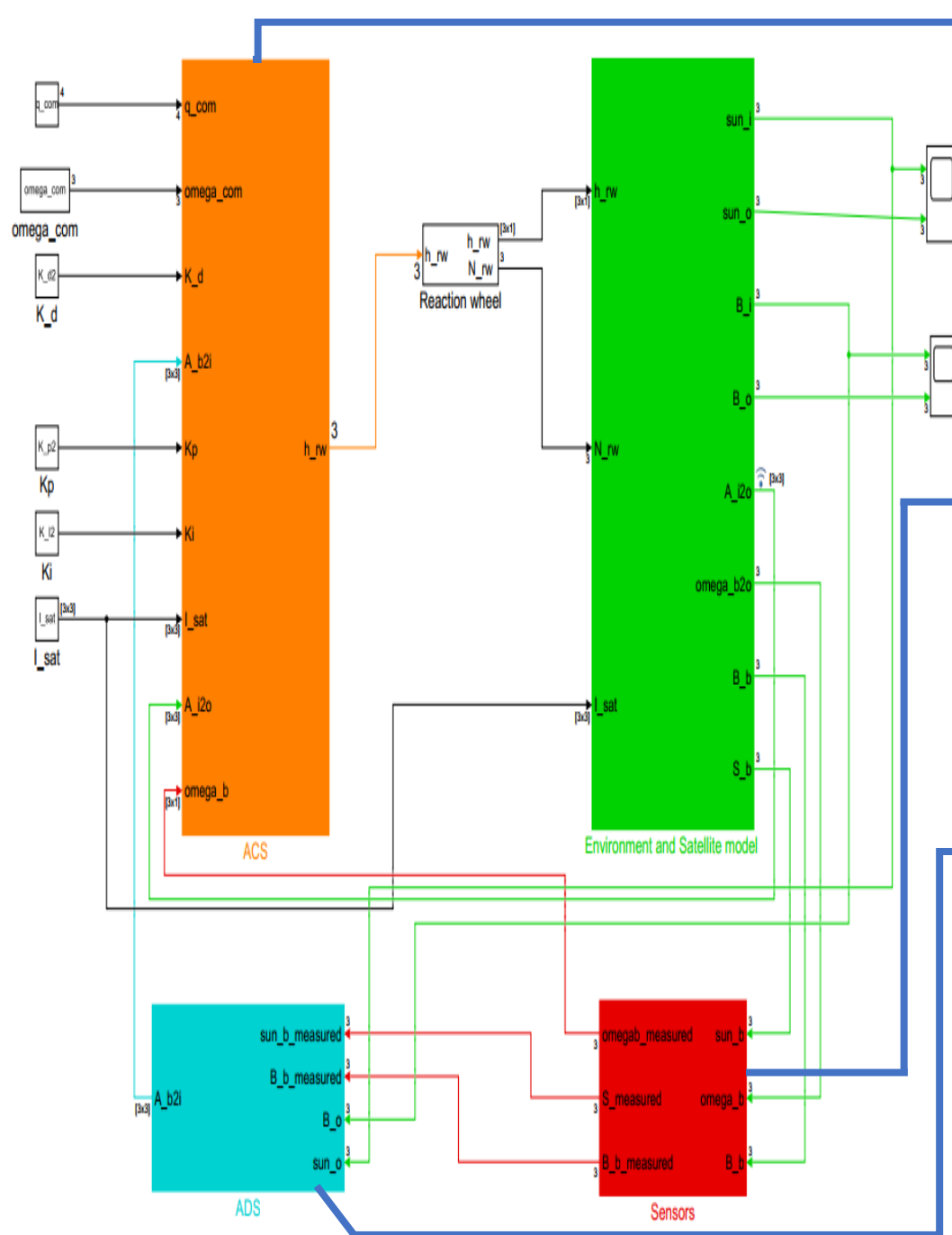


4.2. Overall Configuration

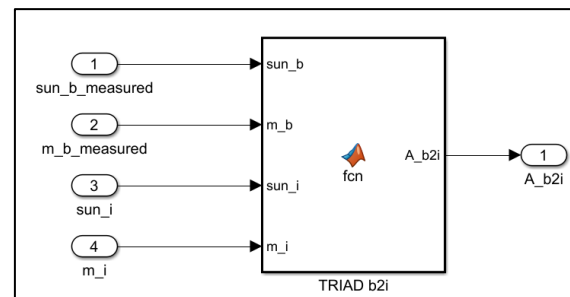
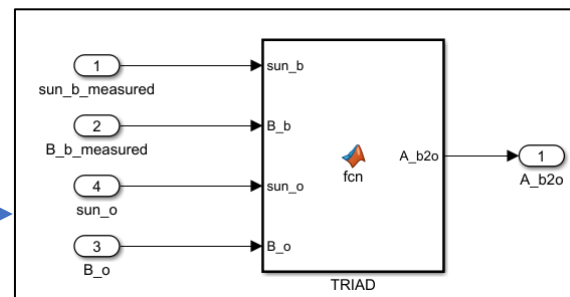
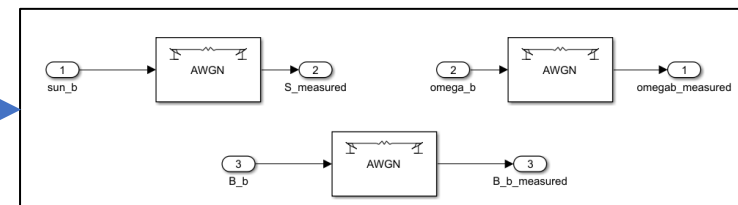
The models includes these following subsystems:

- Attitude Control System (ACS)
- Attitude Determination System (ADS)
- Environment and Satellite model
- Sensors
- Reaction wheels





$$\mathbf{N}_{RW} = K_P \mathbf{I} \dot{\mathbf{q}}_{err} + K_D \mathbf{I} \dot{\hat{\omega}}_b^o - \hat{\omega}_b^i \times (\mathbf{I} \hat{\omega}_b^i + \mathbf{h}_{RW})$$



Attitude Determination System

$$\bar{\mathbf{o}}_1 = \bar{\mathbf{B}}_b, \quad \bar{\mathbf{o}}_2 = \bar{\mathbf{B}}_b \times \bar{\mathbf{S}}un_b, \quad \bar{\mathbf{o}}_3 = \bar{\mathbf{o}}_1 \times \bar{\mathbf{o}}_2$$

$$\bar{\mathbf{r}}_1 = \bar{\mathbf{B}}_o, \quad \bar{\mathbf{r}}_2 = \bar{\mathbf{B}}_o \times \bar{\mathbf{S}}un_o, \quad \bar{\mathbf{r}}_3 = \bar{\mathbf{r}}_1 \times \bar{\mathbf{r}}_2$$

$$\mathbf{A}_b^o = [\bar{\mathbf{o}}_1 \ \bar{\mathbf{o}}_2 \ \bar{\mathbf{o}}_3][\bar{\mathbf{r}}_1 \ \bar{\mathbf{r}}_2 \ \bar{\mathbf{r}}_3]^T$$

Attitude Control System

Sensors

Environment and Satellite model

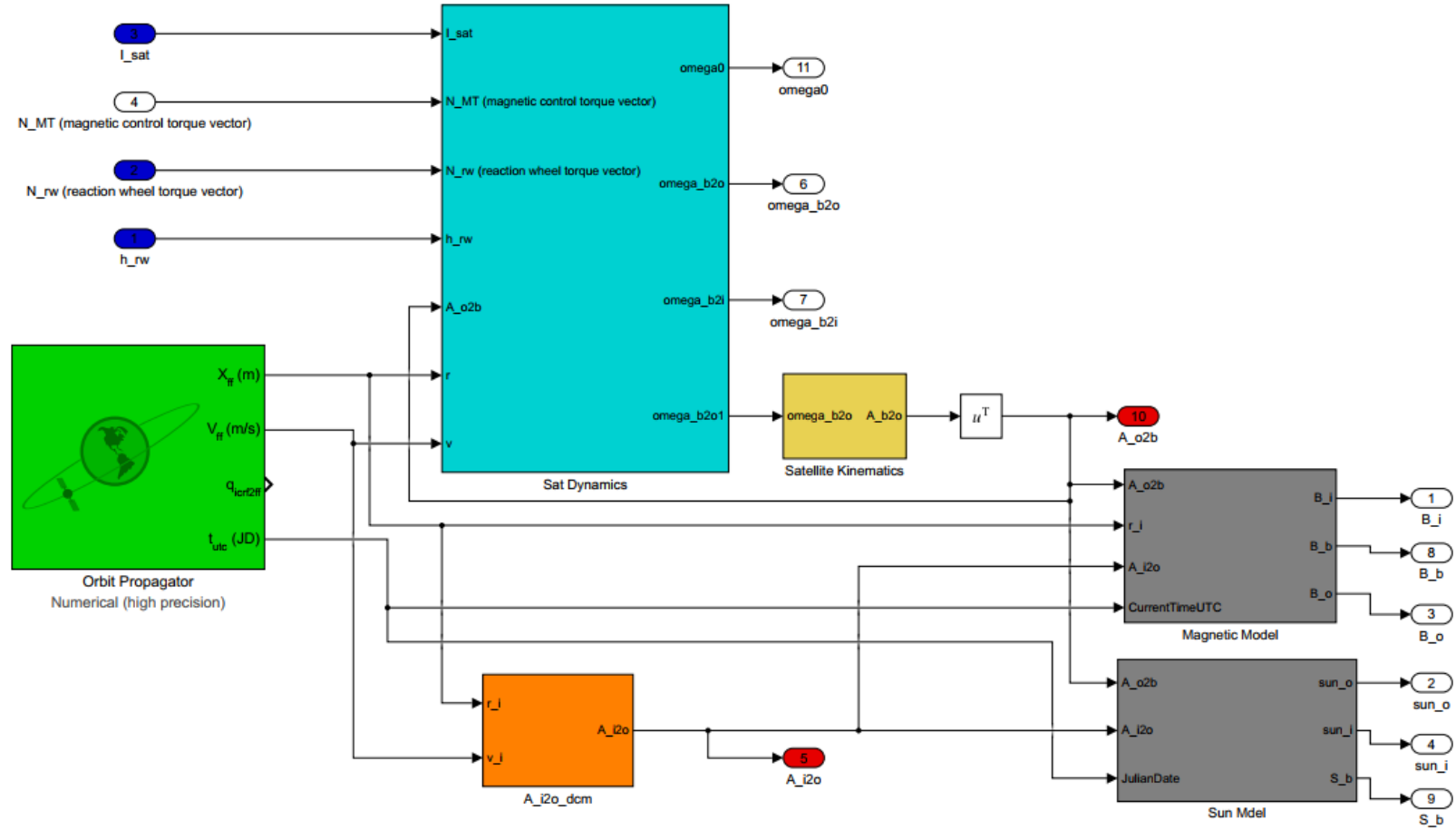
The Environment and satellite model includes the following blocks:

- Orbit Propagator
- Satellite Dynamics
- Satellite Kinematics

$$\mathbf{I}\dot{\boldsymbol{\omega}}_{\mathbf{B}}^{\mathbf{I}} = \mathbf{N}_{GG} + \mathbf{N}_D + \mathbf{N}_{RW} + \mathbf{N}_{MT} - \boldsymbol{\omega}_{\mathbf{B}}^{\mathbf{I}} \times (\mathbf{I}\boldsymbol{\omega}_{\mathbf{B}}^{\mathbf{I}} + \mathbf{h}_{RW})$$

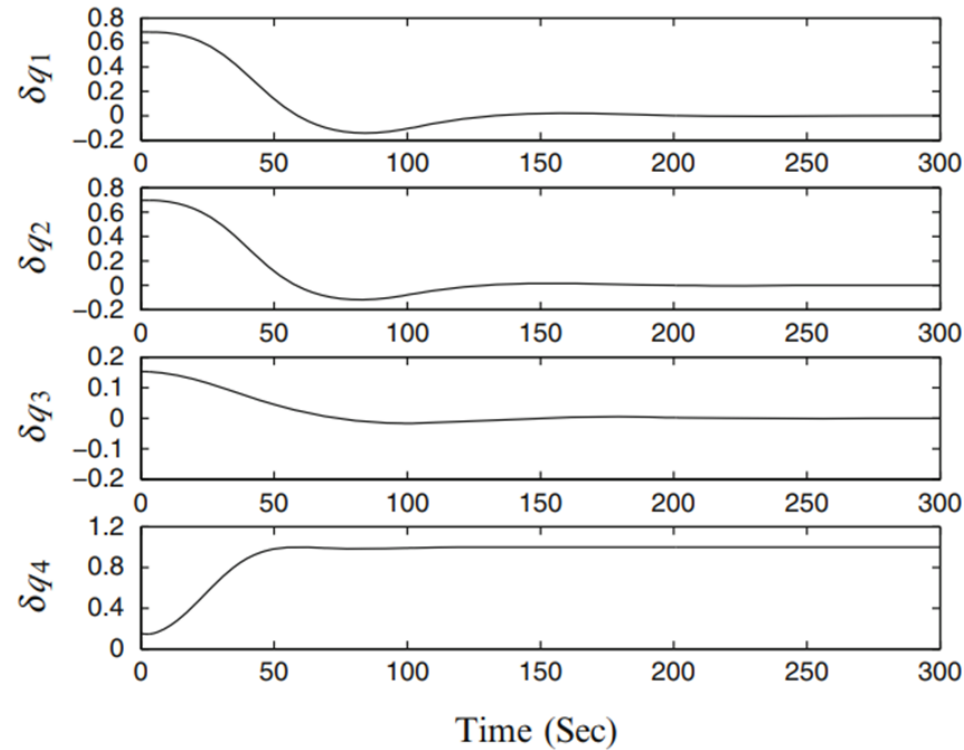
- Magnetic Model
- Sun Model

$$\begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \\ \dot{q}_4 \end{bmatrix} = 0.5 \begin{bmatrix} 0 & \omega_{zo} & -\omega_{yo} & \omega_{xo} \\ -\omega_{zo} & 0 & \omega_{xo} & \omega_{yo} \\ \omega_{yo} & -\omega_{xo} & 0 & \omega_{zo} \\ -\omega_{xo} & -\omega_{yo} & -\omega_{zo} & 0 \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{bmatrix}$$

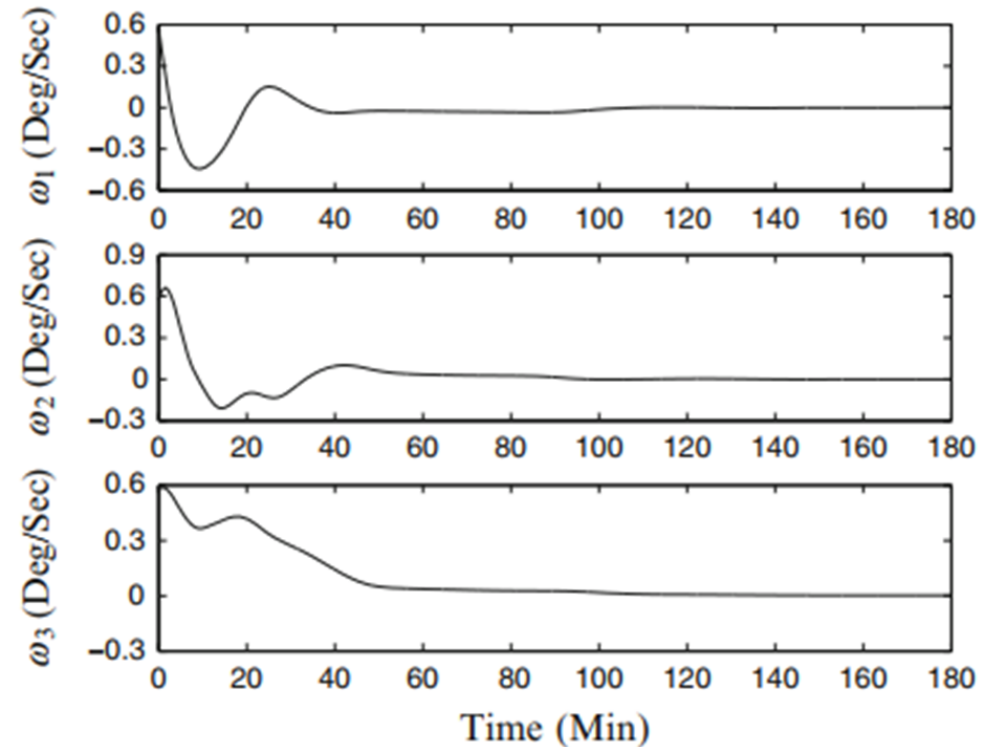


5. Results

5.1. Expected Outcomes



Expected quaternion error with
 $q_{ini} = [0.6853, 0.6953, 0.1531, 0.1531]$

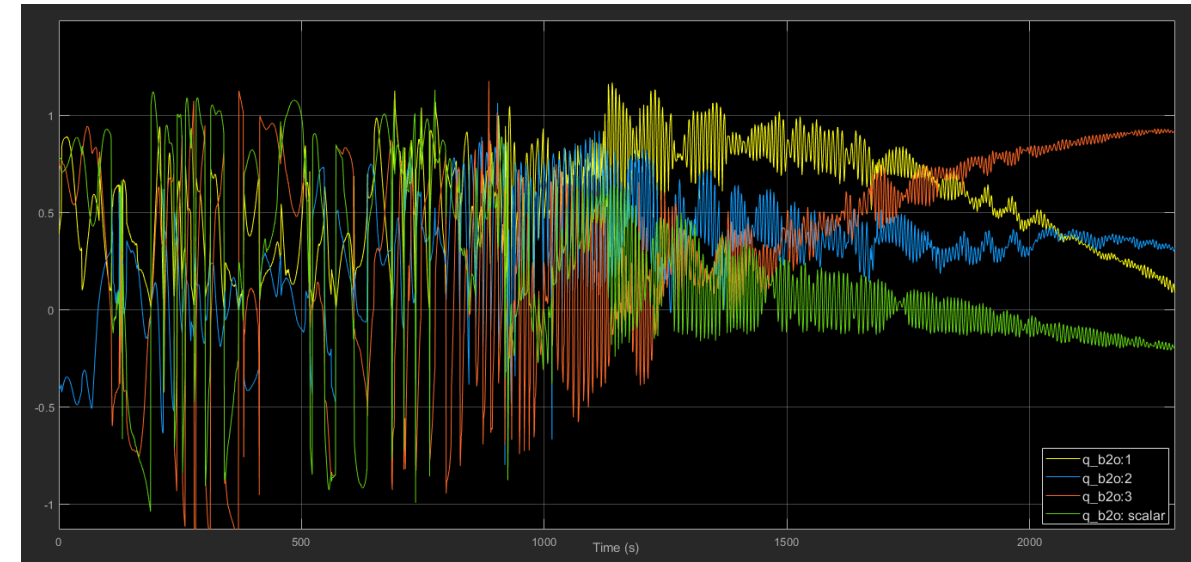


Expected angular velocity with
 $\omega_{ini} = [0.6, 0.6, 0.6]$ degree/sec

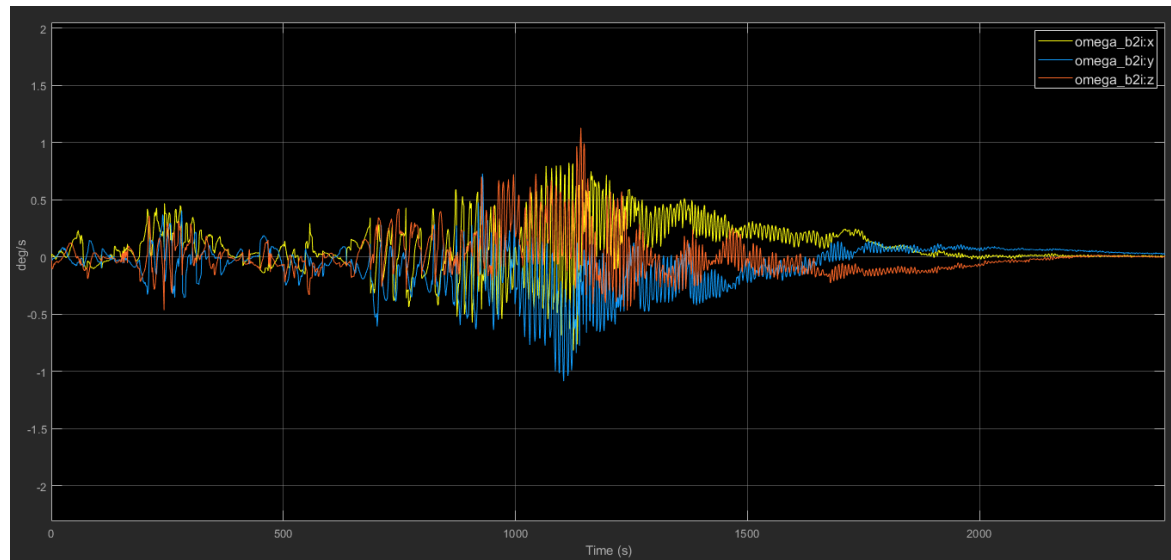
5.2. Outcome from the built model

Parameter	Description	Value	Unit
q_ini	Initial value of quaternion q_b^o	[0,0,0,1]	None
q_com	Commanded quaternion q_b^o	[0.6853,0.6953,0.1531,0.1531]	None
omega_ini	Initial angular velocity ω_b^o	[0,0,0]	degree/second
omega_com	Commanded tracking angular velocity ω_b^o	[0,0,0]	degree/second
K_RW	Reaction Wheel gain	0.15	None
K_D	Derivative gain	5.25	None
K_P	Proportional gain	22	None
I_sat	Inertia matrix of the CubeSat	[0.0011, 0, 0; 0, 0.0012, 0; 0, 0, 0.0014]	$\text{kg} \times \text{m}^2$
h_rw_ini	Initial angular momentum of the reaction wheels	[0,0,0]	$\text{kg} \times \text{m}^2/\text{s}$

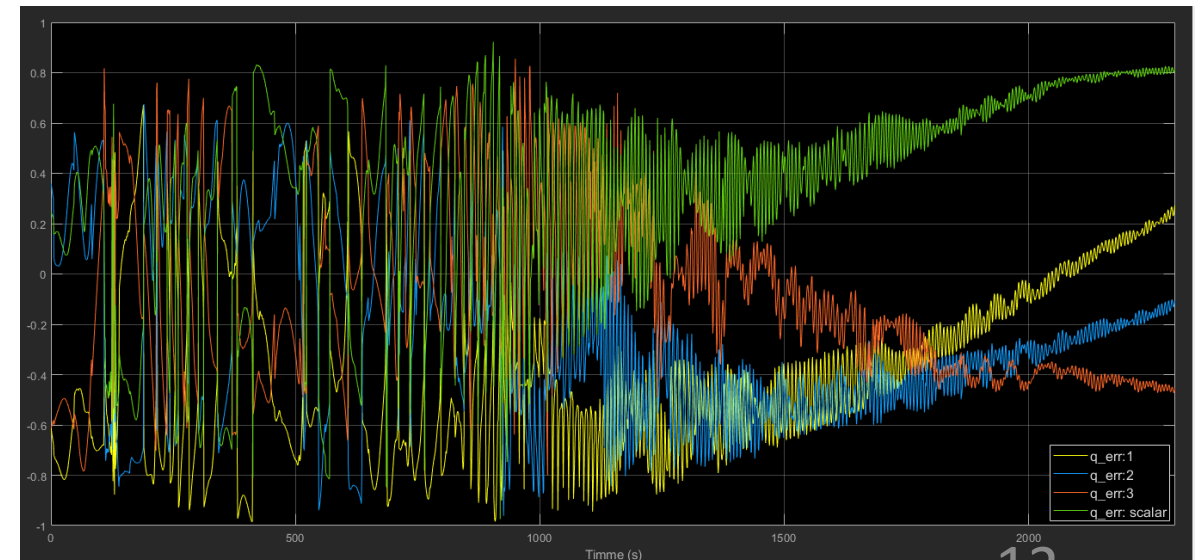
TRIAD b2i



Quaternion q_b2o



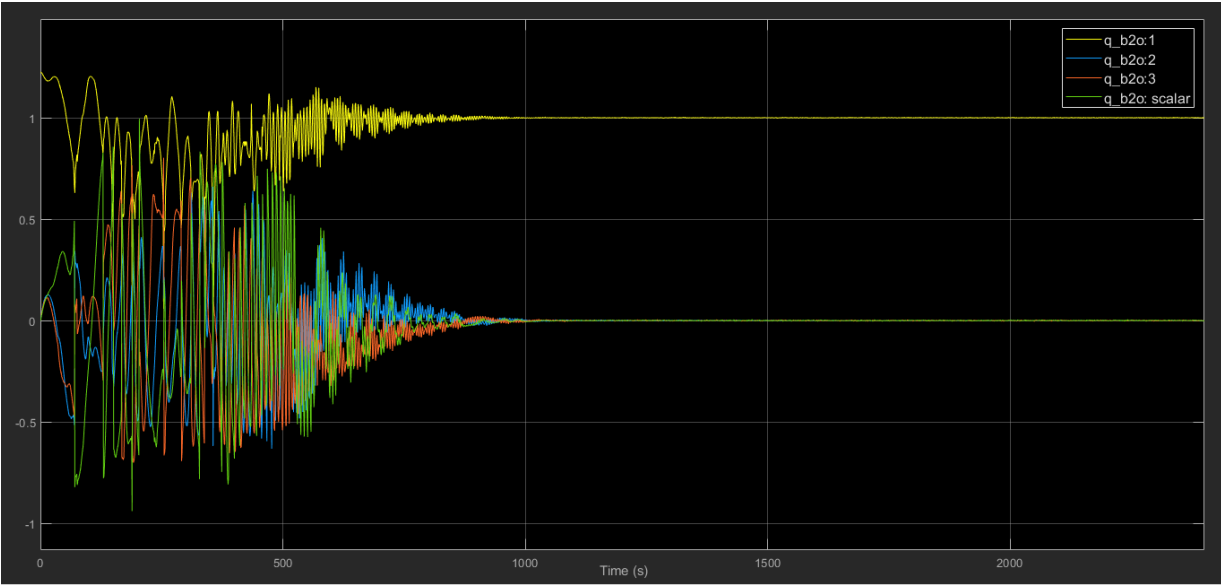
Angular velocity



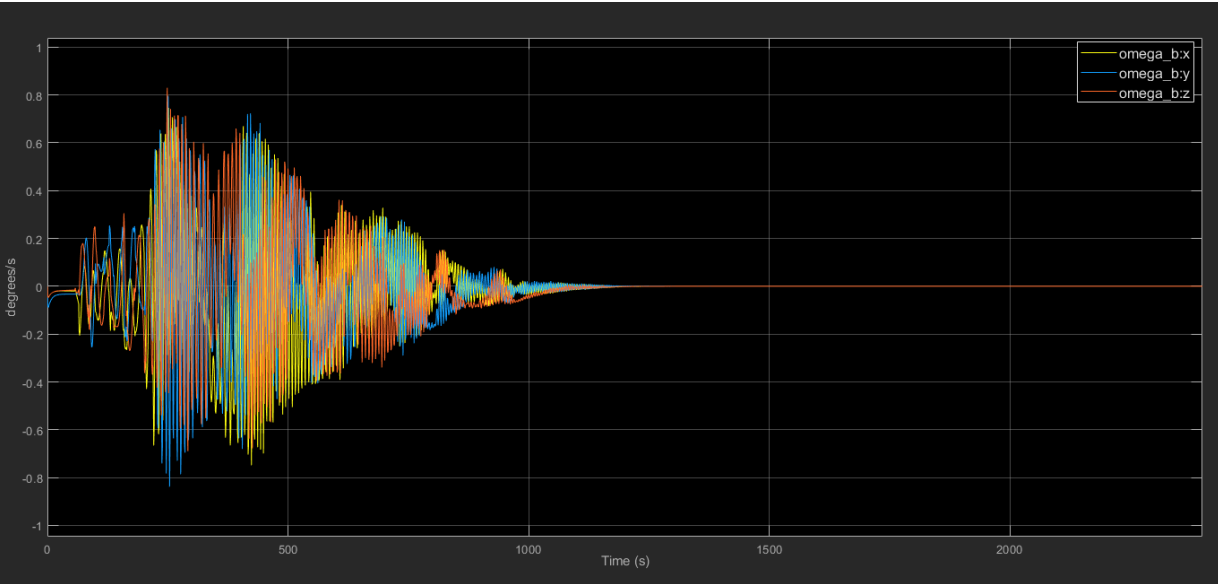
Quaternion error

TRIAD b2o

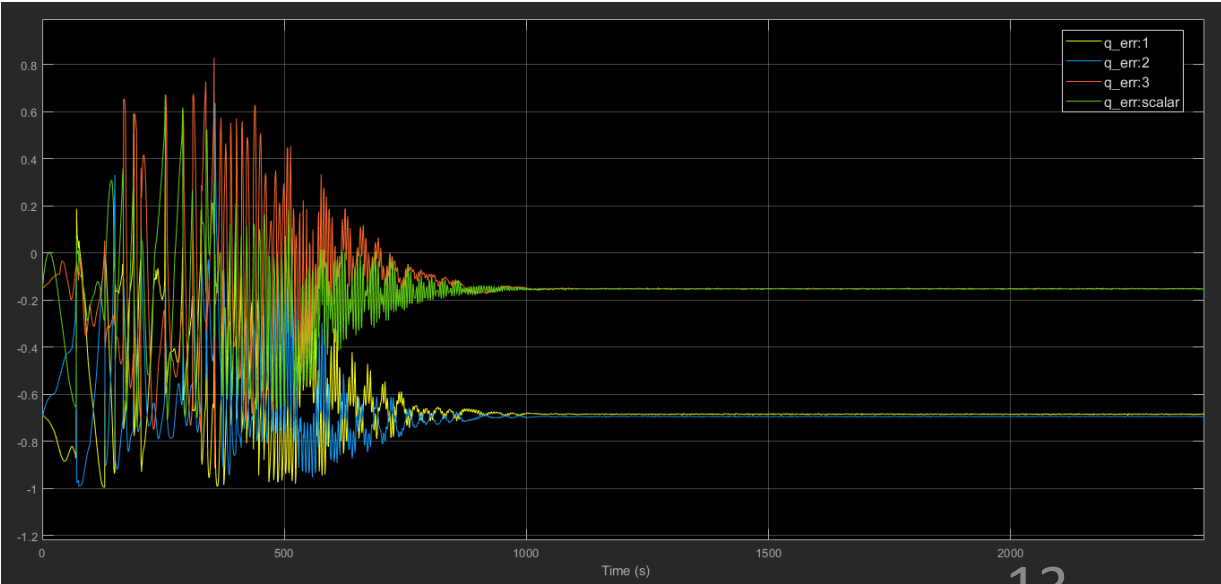
Parameter	Description	Value	Unit
q_ini	Initial value of quaternion q_b^o	[0,0,0,1]	None
q_com	Commanded quaternion q_b^o	[0.6853,0.6953,0.1531,0.1531]	None
omega_ini	Initial angular velocity ω_b^o	[0,0,0]	degree/second
omega_com	Commanded tracking angular velocity ω_b^o	[0,0,0]	degree/second
K_RW	Reaction Wheel gain	0.15	None
K_D	Derivative gain	5.25	None
K_P	Proportional gain	22	None
I_sat	Inertia matrix of the CubeSat	[0.0011, 0, 0; 0, 0.0012, 0; 0, 0, 0.0014]	$\text{kg} \times \text{m}^2$
h_rw_ini	Initial angular momentum of the reaction wheels	[0,0,0]	$\text{kg} \times \text{m}^2/\text{s}$



Quaternion q_b2o



Angular velocity



Quaternion error

6. Conclusions

We have build a ADCS model on CATIA V5 and MATLAB/Simulink. This simulation has all the main functions of an ADCS: the ADS, ACS, Sensors, Reaction Wheels and Environment model. All the blocks can operate well if separated and can return the results when integrated into a complete model.

The output showed that the simulation worked, however, the outcomes of these blocks are erroneous, with large errors and unsuccess in achieving the ADCS requirements stated previously.

The errors can come from:

- Not well-defined parameters in the model
- Coordinates are not perfectly transformed
- Some mistakes in the model

Plans for the future:

- Continue to perfect the model
- Ask experts for modifications
- Try with different coordinate frames and attitude representations
- May add some other environmental effects in the environment model.

7. References

Pelton, J. N., & Madry, S. (2020). *Handbook of Small Satellites: Technology, Design, Manufacture, Applications, Economics and Regulation*. Springer.

Markley, F. L., & Crassidis, J. L. (2014). *Fundamentals of spacecraft attitude determination and control* (pp. 361-364). New York, NY, USA:: Springer New York.

Chen, X., Steyn, W., & Hashida, Y. (2000, August). *Ground-target tracking control of earth-pointing satellites*. In AIAA Guidance, Navigation, and Control Conference and Exhibit (p. 4547).

Le, H.,(2020). *Lectures in "Space System Design" module*, University of Science and Technology of Hanoi.

Thank you for listening!

